What a great coach knows — and what a workout-suggesting AI must know — before writing a single session

This guide is the decision layer. The companion spreadsheet holds 2,000+ classified exercises, stretches, warm-ups, cardio, equipment maps, and form-error rules. This document is the brain that decides which of those rows a human actually gets.

1. The job is not ‘pick exercises’

A competent coach does not start with ‘Monday is chest.’ They start with a person: a goal, a body, a calendar, a room full of (or empty of) equipment, a medical history, and a nervous system that will only adapt if it can recover. National credentials that serious coaches study — ACSM, NASM, ACE, NSCA — all put assessment and program design above exercise trivia. An AI that only has an exercise database will hallucinate confidence. An AI that has screening, contraindications, pattern regressions, load math, and a referral rule can be a useful assistant.

2. Knowledge stack (what top coaches actually study)

No single weekend course is enough. The craft is stacked. Flagship credentials and why they exist:

Body of knowledge	What it puts in the coach’s head
ACSM Guidelines + CPT/EP-C	Preparticipation screening (PAR-Q+), FITT-VP, special populations, exercise-is-medicine discipline. The evidence floor.
NASM CPT + OPT + CES	Human movement science, postural distortion patterns (upper/lower crossed, pronation), the 5-phase OPT model, inhibit-lengthen-activate-integrate.
ACE CPT + IFT + behavior	Integrated Fitness Training model and the psychology layer: stages of change, motivational interviewing tone, adherence.
NSCA CSCS / CPSS	Needs analysis, periodization, energy systems, testing, Olympic/speed/agility, athlete monitoring. The S&C; standard.
USAW / StrongFirst / CrossFit L1	Skill standards for barbell Olympic lifts, kettlebell hardstyle, mixed-modal templates. Mechanics before intensity.
FMS / FRC / DNS / McGill / PRI	Screens and high-resolution movement. Not mandatory, but this is what 10-year coaches add.
Precision Nutrition L1	Nutrition coaching (not dietetics). Hand portions, environment design, scope of practice.
CPR/AED + legal + notes	Emergency action plan, informed consent, SOAP notes, insurance. Non-negotiable.

Textbooks that sit behind those courses: NASM Essentials of Personal Fitness Training; ACSM’s Guidelines for Exercise Testing and Prescription; NSCA’s Essentials of Strength Training and Conditioning; McGill’s Back Mechanic / Ultimate Back Fitness; Cook’s Movement; Verkhoshansky/Siff Supertraining for historical periodization literacy; Issurin on block periodization.

3. Gates before a workout is allowed to exist

If an AI cannot complete this list, it should ask questions — not invent a 6-day bro split.

- **Goal in 12 weeks.** Strength, hypertrophy, fat loss while keeping muscle, pain-free function, aerobic base, sport. One primary adaptation.
- **Training age and schedule.** Days, minutes, travel. The honest constraint beats the fantasy constraint.
- **Equipment reality.** Mat / bands / one kettlebell / pair of DBs / full gym. Write to the room.
- **PAR-Q+ / red flags.** Chest pain, unexplained dyspnea, dizziness, night pain, radiating numbness, post-op, pregnancy complications, uncontrolled medical disease → refer, do not intensify.
- **Pain map.** Where, when, which movement, sharp vs dull. Coaches do not diagnose. They regress and refer.
- **Movement snapshot.** Can they squat to a box with heels down? Hinge without lumbar flexion? Reach overhead without rib flare? Row without shrugging?
- **Recovery capacity.** Sleep, kids, shift work, already-running 50 km weeks, caloric deficit. Volume lives here.
- **Preference.** Adherence is a training variable. The 'perfect' program they skip is a zero.
- **Stop rules.** Sharp or radiating pain, chest pain, sudden unilateral weakness — session over.

4. The science of suggesting the workout

4.1 Principles that do not expire

- **SAID.** Specific Adaptation to Imposed Demands. Heavy slow squats build squat strength, not a 40-yard dash.
- **Progressive overload.** Load, reps, sets, range, density, or skill. Change one thing at a time for beginners.
- **Specificity.** Hypertrophy, max strength, and VO2max are different adaptations. Name the one you want.
- **Individuality + reversibility.** Responders vary. Unused qualities fade. Maintenance doses exist.
- **FITT-VP (ACSM).** Frequency, Intensity, Time, Type, Volume, Progression. If you cannot fill those six blanks, you do not have a program.
- **Fitness-fatigue.** The session is the stimulus. The adaptation is after. Deloads drop fatigue faster than fitness.
- **Pattern before load, load before fatigue, fatigue before chaos.** Do not add intensity to a pattern the person does not own.

4.2 The NASM OPT model (commercial-PT default)

Three levels, five phases: (1) Stabilization Endurance — 12–20 slow reps, lower load, own the pattern; (2) Strength Endurance — strength lift paired with a stability move; (3) Hypertrophy / muscular development — 6–12 reps, higher volume; (4) Maximal Strength — 1–5 reps, high load, long rest; (5) Power — high-load strength paired with explosive intent. You do not start a desk-stiff beginner in phase 4. You do not train kipping pull-ups on a shoulder that cannot depress the scapula.

4.3 Practical volume and intensity landmarks

Quality	Useful starting landmark
Hypertrophy volume	About 10–20 hard sets per muscle per week for intermediates; beginners grow on less.
Strength intensity	Most work at RPE 6–8 (2–4 reps in reserve). True 1RMs are rare and expensive.
Frequency	Each muscle / pattern 2–3 times per week beats one heroic session for most humans.
Rest	Strength 3–5 min; hypertrophy 1–3 min; metabolic work 30–90 s.
Aerobic default	Build toward 150–300 min moderate or 75–150 vigorous / week (ACSM population target), individualized from the current base.
Warm-up	3–5 min pulse raise → 3–6 drills that look like today's patterns → 1–3 ramp-up sets.
Deload	Every 4–8 weeks, or when sleep/pain/motivation fall over: cut volume ~40–60%.

4.4 Energy systems (so you do not stack stupid)

Phosphagen (≤10 s, long rest) is max strength and short sprints. Glycolytic is 20–120 s hard efforts. Oxidative is the easy conversational work that builds the base most adults are missing. A heavy squat day, a 5K time trial, and a fight class in the same 24 hours is not ‘work ethic.’ It is noise. Pick the cost you are willing to pay.

5. Exercise selection rules

Choose **movement patterns first**, named lifts second. The human movement catalog that covers a life is small: squat, hinge, lunge / split stance, horizontal push, vertical push, horizontal pull, vertical pull, carry, core (anti-extension, anti-rotation, anti-lateral flexion, then dynamic), locomotion, and — when earned — plyometric / Olympic.

Then filter the catalog: equipment the person actually has → level they can perform without the compensation pattern → contraindications (no behind-the-neck pressing for a cranky shoulder, no loaded spinal flexion for a flexion-intolerant back unless a clinician is driving) → preference. Every chosen lift must have a **regression** (box squat, landmine press, assisted pull-up, DB RDL) and a **stop rule** from the form-error sheet.

Muscle targeting is real but not mystical. A barbell row hits lats, mid-back, rear delts, biceps, and spinal erectors. If the user yanks with the arms and shrugs, it becomes an ugly biceps/upper-trap exercise. That is why the spreadsheet stores both the intended prime mover and the muscles that steal the rep when posture collapses.

6. Warm-ups, stretches, cardio — different jobs

- **Warm-up** raises tissue temperature, rehearses today’s pattern, and potentiates. Dynamic. Specific. Short. Cat-cow, world’s greatest stretch, glute bridge, wall slides, band pull-aparts, ankle rocks, empty-bar squat.
- **Activation** is low-load work for muscles that are supposed to fire and often don’t (glute med, lower trap, serratus, deep neck flexors, TVA).
- **Static stretching** belongs after training or in a separate session for short, overactive tissue (hip flexors, pec minor, upper trap, calves). Do not stretch a muscle that is already long and weak and calling that ‘mobility.’
- **Cardio** is an adaptation of its own. Zone-2 / talk-test work is the default adult builder. Intervals are a spice after a base exists. Steps and easy walks count.

Corrective logic in one line (NASM CES): inhibit what is overactive → lengthen it → activate what is underactive → integrate it into a standing compound pattern. Foam-rolling a weak glute and then stretching it again is not a program.

7. Form errors are pattern errors

Most ‘bad form’ is one of three things: the setup was wrong, the load was too heavy, or fatigue changed the rep. The body will finish the set with whoever is available. Typical thefts:

Pattern	What it looks like when it breaks	Who stole the rep
Squat	Knees cave, heels lift, torso dumps, butt wink	Adductors, lumbar erectors; quads / glutes clock out
Hinge	Lumbar round, bar drifts, squat-the-hinge, lean-back lockout	Erectors and upper traps; hamstrings unload
Horizontal push	Elbows at 90°, shrug, rib flare, bounce	Anterior delts and upper traps; pecs under-loaded
Vertical push	Low-back arch, chin poke, press in front of the face	Lumbar erectors + upper traps
Row / pull-up	Swing, shrug, chin poke, biceps-only	Upper traps and arms; mid-back silent
Core crunch / raise	Yank the neck, swing the hips	Iliopsoas and cervical flexors
Lunge	Valgus, short step, torso collapse	Adductors and low back; glute med silent

Cue sparingly. One or two cues beat twelve. ‘Ribs down, push the floor, elbow to the hip’ is a coaching session. A paragraph of anatomy during the set is not.

8. Skills an AI must have (the competency model)

Treat this as a hiring spec. If a system cannot do these, it should not be allowed to auto-publish a week of training.

- Intake and red-flag screening with a referral action, not a disclaimer buried under a generated split.
- Needs analysis: goal × constraints × equipment × training age × recovery.
- Movement-pattern literacy and a regression/progression tree for every pattern.
- Anatomy at the level of joint actions, planes, agonist/synergist/antagonist/stabilizer — not a recitation of origins.
- Load math: RPE/RIR or %1RM, weekly hard-set targets, rest, frequency.
- Periodization at least at the level of ‘accumulation / intensification / deload’ or OPT phases.
- Energy-system literacy so hard intervals and heavy hinges are not stacked by accident.
- Special-population rules (pregnancy, older adult power + balance, hypertension, diabetes, osteoporosis, flexion-intolerant backs) and the humility to stay inside scope.
- Behavior-change tone: smallest program they will actually do. Ask what they hated last week.
- Equipment rewriting: the same program expressed for a mat, a hotel DB pair, a kettlebell, and a full gym.
- A closed loop: log last session’s pain, RPE, completed reps, and sleep, then change tomorrow.
- Honesty: no invented 1RMs, no 10-kg-in-10-days promises, uncertainty shown when evidence is thin.

9. A default decision tree

1) Screen. 2) Name the adaptation. 3) Name the days and the room. 4) Pick a template (full-body 3×, upper/lower 4×, kettlebell minimalist, travel mat, in-season maintenance). 5) Fill each day with patterns, then specific catalog lifts that match gear and level. 6) Add a short specific warm-up. 7) Add 1–2 stretches only for tissues that are actually short. 8) Add easy aerobic work on the calendar, not as punishment. 9) Write the stop rules. 10) Review in 4 weeks against sleep, pain, adherence, and a simple performance marker (more reps at the same load, faster easy pace, belt notch).

When in doubt, write a boring full-body session the person can repeat for twelve weeks: squat or split squat, hinge, push, pull, carry, a brace drill, a walk. Progress the load. That program has created more useful humans than any novelty circuit.

10. Scope, ethics, and what this file is not

This package is an educational knowledge base. It is not a diagnosis, not a rehab protocol, not a meal plan, and not a license. Exercise names and classifications were merged from public exercise catalogs (including the public-domain Free Exercise DB) and then labeled with coaching fields — setting, pattern, form-error model, warm-up role — synthesized from ACSM, NASM, ACE, and NSCA teaching models. Individual pain and medical conditions need a human clinician. Individual high-load technique needs a human coach who can see the bar path.

Use with Complete_Coach_Fitness_Knowledge_Base.xlsx. Filter sheet 01 by Setting and Primary_muscle. Read sheets 09–12 before you let any model print a week of work.